

Exact Finite-State Coverage of Hellman and Distinguished-Point Time–Memory Tradeoffs: A Reproducible Study on Reduced-DES Mappings

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Abstract

Cryptanalytic time–memory tradeoffs represent a search space by iterating a fixed mapping and storing selected chain endpoints. Their effective coverage is therefore governed by functional-graph coalescence rather than by independent occupancy.

We present an exact finite-state, reproducible study of Hellman and bounded distinguished-point tables for reduced mappings derived from DES. The artifact contains independent Java and Rust implementations, a literal shared 30-trial workload, complete finite-map enumeration, and matched random-mapping controls. The Java and Rust deterministic outputs are byte-for-byte identical.

For the motivating $N = 2^{16}$ Hellman configuration with $m = 1024$ and $t = 64$, the established Ma–Hong random-function recurrence predicts 16.651% coverage. The measured medians are 16.840% for the DES-derived mapping and 16.810% for matched random mappings, showing that the earlier independent-occupancy reference $1 - e^{-1}$ is not the appropriate Hellman baseline. Parameter sweeps likewise show close agreement between the concrete mapping, random-map controls, and fixed-map coverage theory.

For distinguished points, exact reverse reachability over all 2^{16} states explains why truncation need not vanish as the chain bound grows: trajectories whose eventual functional-graph components avoid the distinguished set can never terminate at a distinguished point. The experiments thus connect classical TMTO theory with exact finite functional graphs and a fully reproducible cryptographic implementation artifact.

1 Introduction

A time–memory tradeoff (TMTO) moves work from the online phase of a search problem into a reusable preprocessing phase. In Hellman’s classical construction [Hel80], a large key or state space is represented by iteration chains while only chain endpoints are retained. The construction therefore occupies a middle ground between exhaustive search, which uses little memory and large online time, and complete table lookup, which uses large memory and very small online time.

The central implementation issue is that a chain table does not contain mt independent samples. Once two iterates of the same mapping meet, their future trajectories coincide; a chain can also revisit one of its own previous states. Consequently, the effective coverage of a table is the number of *distinct* represented states, not its nominal number of positions. This phenomenon is part of the classical TMTO literature rather than a new observation: Hellman’s analysis, later work on random mappings, success probability, perfect tables, false alarms, and rainbow reductions all treat different aspects of collisions, merging, and coverage [Hel80, FO90, Oec03, AJO08, MH09, Hon08].

The purpose of this paper is therefore narrower and experimentally explicit. We ask how accurately established random-function TMTO theory describes a *concrete finite cryptographic mapping* when every represented state can be enumerated exactly, and whether independent implementations reproduce the same semantics and finite-state statistics.

Concrete iteration map. For a reduced state x , a canonical DES-key encoding $K(x)$, a fixed plaintext P , encryption E , and reduction R , we study

$$F(x) = R(E_{K(x)}(P)).$$

DES is obsolete and is used only as a compact deterministic fixture. The state dimension is deliberately reduced so that exact coverage, cycles, endpoint multiplicities, and recovery behavior can be computed rather than estimated from sampling.

A motivating discrepancy and its correct baseline. The repository’s initial $N = 2^{16}$ experiment used $m = 1024$ fixed-length chains of length $t = 64$, so $mt = N$. Treating the mt chain positions as independent samples would give the occupancy expression

$$1 - e^{-mt/N} = 1 - e^{-1} \approx 63.2\%.$$

That number is *not* the expected coverage of a Hellman matrix. Using the random-function recurrence of Ma and Hong [MH09], the same parameters predict approximately 16.651% coverage of the state space. The measured median exact coverage is approximately 16.8%. Thus the initial measurement is in close agreement with established random-function theory.

This distinction also exposes an important parameterization point. The standard Hellman matrix-stopping regime is expressed in terms of mt^2 relative to N , not merely mt/N . Our initial configuration satisfies $mt = N$ but $mt^2 = 64N$; it should therefore be treated as a finite-state stress point, not as a standard matrix-stopping configuration.

Contributions and scope. This paper is an experimental and reproducibility study, not a claim to have discovered chain coalescence or classical TMTO coverage theory. Its intended contributions are:

1. a precise, executable reduced-DES state model with canonical DES parity handling and an explicit reduction;
2. exact coverage and exact-recovery instrumentation for Hellman and bounded distinguished-point tables;
3. direct comparison of concrete cryptographic mappings with established random-function coverage predictions over controlled finite-state parameter sweeps;
4. independent Java and Rust reference implementations, with shared semantic fixtures and, before any timing comparison, a common committed experiment plan;
5. a reproducibility artifact in which numerical claims are generated from committed inputs, outputs, scripts, and environment metadata.

We make no claim to break full-size DES or AES, and no implementation in this repository is intended for production cryptography.

2 Related Work

Classical time–memory tradeoffs. Hellman introduced the cryptanalytic time–memory tradeoff for block-cipher key recovery and analyzed the effect of chain collisions under random-function assumptions [Hel80]. Fiat and Naor later gave rigorous time/space tradeoffs for function inversion beyond the random-function setting [FN91]. Barkan, Biham, and Shamir developed a general model of cryptanalytic TMTOs based on stateful random graphs and proved rigorous bounds covering classical and rainbow-style constructions [BBS06]. Recent work continues to revisit the fundamental limits of preprocessing and function inversion [GGPSD23].

Random mappings and coverage. Iteration chains naturally induce functional graphs. Flajolet and Odlyzko systematically analyzed random-mapping statistics including cycles, components, and related structural parameters [FO90]. For Hellman tables specifically, Ma and Hong studied the expected coverage rate and supplied a recurrence that closely matches experiments under a random function model [MH09]. Their framework is a central baseline for the present paper: our goal is not to replace it, but to test it against exact coverage of a concrete finite cryptographic mapping.

Rainbow and perfect tables. Oechslin’s rainbow construction changes the reduction across chain positions, reducing the conditions under which collisions turn into permanent chain merges [Oec03]. Avoine, Junod, and Oechslin analyzed perfect classic, distinguished-point, and rainbow tables, including practical performance considerations [AJO08]. These works make clear that coverage, preprocessing cost, and lookup cost must be compared at matched success probabilities rather than through nominal table sizes alone.

False alarms and distinguished points. Endpoint matches do not always imply successful recovery: false alarms can force regeneration of candidate chains and materially affect the online cost. Hong analyzed this cost for Hellman and rainbow tradeoffs [Hon08]. Lee and Hong compared major perfect-table variants and developed detailed analysis for distinguished-point constructions [LH12].

Position of this work. The present paper focuses on reproducibility at finite state sizes where the entire mapping and table coverage can be enumerated. This permits a direct three-way comparison among established random-function predictions, a concrete DES-derived mapping, and independent Java/Rust implementations. The value of the study is therefore in exact instrumentation, controlled comparisons, and reproducible validation—not in reintroducing known TMTTO phenomena.

3 Model and Definitions

This section fixes the finite-state model used by both implementations and by all exact-coverage measurements. The definitions deliberately separate the cryptographic primitive from the reduction and from the TMTTO table structure.

3.1 Reduced effective DES state space

For an integer

$$1 \leq b \leq 28,$$

let

$$X_b = \{0, 1, \dots, 2^b - 1\}, \quad N = |X_b| = 2^b.$$

The parameter b counts *effective DES key bits*; it is not a count of raw bits in the eight-byte DES key representation.

DES stores eight bytes, with one parity bit in each byte. The experiment keeps the first four bytes fixed and allows the final four bytes to carry at most $4 \cdot 7 = 28$ effective data bits. Let

$$K_b : X_b \longrightarrow \{0, 1\}^{64}$$

denote the canonical encoding used by the artifact. The state is decomposed into four seven-bit groups, placed into the four variable DES bytes from most significant group to least significant group, and the least-significant bit of each byte is then chosen to give odd DES parity. Unused high state bits are zero. This convention makes the state-to-key map deterministic and invertible on its image.

The fixed four-byte prefix uses seven effective one-bits in each byte. Its purpose is solely to preserve the effective-data structure of the historical prototype; it is not a security parameter.

3.2 Encryption oracle and reduction

Let P be the fixed ASCII plaintext

HPC reproducibility fixture.

For $x \in X_b$, define the deterministic encryption oracle

$$G_b(x) = \text{DES}_{K_b(x)}(P),$$

using DES in ECB mode with PKCS#5 padding. DES is obsolete and appears here only as a small deterministic fixture whose entire reduced state space can be enumerated.

The reduction

$$R_b : \{0, 1\}^* \longrightarrow X_b$$

uses the final four ciphertext bytes. From each such byte it discards the least-significant bit, retains the upper seven bits, concatenates the resulting four seven-bit words into a 28-bit integer, and finally truncates to the low b bits. Equivalently, if the final four ciphertext bytes are c_0, c_1, c_2, c_3 , interpreted as unsigned integers, define

$$u_i = \lfloor c_i / 2 \rfloor \in \{0, \dots, 127\},$$

$$U = (((u_0 \cdot 2^7 + u_1) \cdot 2^7 + u_2) \cdot 2^7 + u_3),$$

and

$$R_b(c) = U \bmod 2^b.$$

The TMTO iteration map is therefore

$$F_b = R_b \circ G_b : X_b \longrightarrow X_b. \tag{1}$$

For fixed b and P , this is one deterministic functional graph on N vertices.

3.3 Targets and exact recovery

A trial chooses a target state

$$x^* \in X_b$$

and publishes only

$$y^* = G_b(x^*).$$

An algorithm reports exact recovery only when it returns a state $\hat{x}$ such that

$$G_b(\hat{x}) = y^*.$$

The implementation therefore verifies candidate states against the full target ciphertext rather than treating an endpoint collision as success.

For the deliberately reduced state spaces studied here, exhaustive search is a reference method: test $x = 0, 1, \dots, N - 1$ until the target ciphertext is matched. It has exact state-space coverage one and serves as a correctness control rather than as a practical attack on DES.

3.4 Hellman matrix and table

Choose m distinct start states

$$S = (s_1, \dots, s_m) \subset X_b.$$

For a chain length $t \geq 1$, define the Hellman matrix

$$H_{i,0} = s_i, \quad H_{i,j+1} = F_b(H_{i,j}), \quad 0 \leq j < t.$$

Thus row i contains $t + 1$ states

$$H_{i,0}, H_{i,1}, \dots, H_{i,t},$$

but the table represents the first t positions

$$H_{i,0}, \dots, H_{i,t-1}$$

and stores the pair

$$(H_{i,0}, H_{i,t}).$$

This convention matches the coverage definition used by Ma and Hong [MH09] and matches the artifact's exact coverage analyzer.

The implementation indexes all starts by their final endpoint. Multiple rows with the same endpoint are retained rather than silently overwritten.

3.5 Exact Hellman coverage

The represented-state set of one Hellman matrix is

$$\mathcal{C}_H = \{H_{i,j} : 1 \leq i \leq m, 0 \leq j < t\}. \quad (2)$$

Its exact covered-state count and state-space coverage fraction are

$$C_H = |\mathcal{C}_H|, \quad \rho_H = \frac{C_H}{N}.$$

The normalized coverage rate used in the classical literature is instead

$$\text{CR}_H = \frac{C_H}{mt}.$$

The two normalizations answer different questions:

$$\rho_H = \frac{mt}{N} \text{CR}_H.$$

When $mt = N$, as in the initial $b = 16, m = 1024, t = 64$ stress configuration, these two numerical values happen to coincide. They should not otherwise be conflated.

The artifact computes C_H exactly with a bit set over all N reduced states; this analysis is performed outside the timed preprocessing and lookup paths.

3.6 Endpoint collisions and candidate verification

Let

$$e_i = H_{i,t}$$

be the endpoint of row i . The number of distinct stored endpoints is

$$E_H = |\{e_1, \dots, e_m\}|.$$

Endpoint multiplicity is retained because several starts may produce the same endpoint.

During online lookup, the target ciphertext is reduced and projected forward to possible endpoint positions. If a projected value hits the endpoint index, all previously unchecked rows stored under that endpoint are regenerated from their starts. An endpoint hit that fails to reproduce y^* is a false candidate event. We therefore distinguish:

- *projected endpoint matches*: online projections that hit a stored endpoint key;
- *candidate rows checked*: distinct stored rows regenerated after such hits;
- *exact recovery*: regeneration finds a state whose full ciphertext equals y^* .

These counters measure concrete lookup work and should not be replaced by a success/failure bit alone.

3.7 Distinguished-point tables

Fix a difficulty parameter

$$1 \leq d \leq b$$

and define

$$D_d(x) = 1 \iff x \bmod 2^d = 0.$$

Thus a fraction exactly 2^{-d} of the state space is distinguished. Under a uniform-state heuristic, a fresh state is distinguished with probability

$$p_d = 2^{-d},$$

so an unconstrained geometric stopping model has mean

$$1/p_d = 2^d$$

transitions.

For a maximum chain length L , a distinguished-point row starts at s_i and iterates F_b until the first transition whose output is distinguished:

$$\tau_i = \min\{j \in \{1, \dots, L\} : D_d(F_b^j(s_i)) = 1\}.$$

If no such j exists, the row is *truncated* and is not stored. A successful row stores its start and distinguished endpoint

$$(s_i, F_b^{\tau_i}(s_i)).$$

The exact DP coverage used by the artifact is

$$\mathcal{C}_{\text{DP}} = \bigcup_{\tau_i \text{ exists}} \{F_b^j(s_i) : 0 \leq j < \tau_i\}.$$

In particular, truncated rows are excluded from this represented-state set. This convention is made explicit because other implementations may count partially generated truncated rows differently.

3.8 Measured quantities

For each parameter configuration and trial, the study records as available:

state-space size	N ,
exact coverage	C/N ,
exact recovery	$\mathbf{1}[\hat{x} \text{ recovered}]$,
stored rows	number of accepted table rows,
distinct endpoints	E ,
truncated DP rows	T_{trunc} ,
endpoint/projection hits	Q_{end} ,
candidate rows regenerated	Q_{cand} ,
offline time	T_{off} ,
online time	T_{on} .

Later parameter sweeps additionally record functional-graph statistics such as row merges, self-intersections, and cycle/component information. Those quantities are reported only after the corresponding instrumentation is committed.

4 Random-Function Coverage Baselines

The concrete map F_b in Eq. (1) is deterministic. Classical TMTO analyses nevertheless model the underlying iteration function as a random mapping on a set of size N . This section states the random-function baseline against which our exact DES-derived measurements are compared.

4.1 Why independent occupancy is not a Hellman prediction

If q points were sampled independently and uniformly from an N -element set, the expected number of distinct observed states would be

$$N \left(1 - \left(1 - \frac{1}{N} \right)^q \right) \approx N(1 - e^{-q/N}).$$

Putting $q = mt$ gives the familiar occupancy expression

$$\rho_{\text{iid}} \approx 1 - e^{-mt/N}. \quad (3)$$

For $mt = N$, this is $1 - e^{-1} \approx 0.6321$.

Equation (3) is useful as a deliberately naive reference, but Hellman matrix positions are not independent samples. Within each row, later entries are deterministic iterates of earlier entries; once two trajectories meet under a fixed function, their suffixes coincide. Classical Hellman analysis and later random-mapping work explicitly account for this dependence [Hel80, FO90, MH09]. We therefore do not use Eq. (3) as the expected Hellman coverage.

4.2 Ma–Hong expected coverage recurrence

Ma and Hong [MH09] analyze the first t columns of an $m \times (t + 1)$ Hellman matrix. Let

$$\overline{H}_k = \{H_{i,j} : 1 \leq i \leq m, 0 \leq j \leq k\}$$

and let p_k denote, after normalization by N , the expected number of states appearing for the first time in column k . Introducing

$$s_k = \sum_{j=0}^{k-1} p_j,$$

their large- N approximation yields

$$s_0 = 0, \quad s_{k+1} = 1 - \exp\left(-\frac{m}{N}\right) \exp(-s_k). \quad (4)$$

The expected coverage rate relative to the mt nominal matrix positions is

$$\text{ECR}(N, m, t) = \frac{N}{mt} s_t. \quad (5)$$

Consequently, the predicted fraction of the *whole state space* covered by the first t columns is simply

$$\rho_{\text{MH}}(N, m, t) = \frac{mt}{N} \text{ECR}(N, m, t) = s_t. \quad (6)$$

This distinction in normalization is particularly useful in our experiments. The artifact reports C_H/N , whereas much of the TMTO literature reports coverage relative to mt .

4.3 Initial configuration

For

$$N = 2^{16} = 65536, \quad m = 1024, \quad t = 64,$$

Eq. (4) gives

$$s_{64} \approx 0.1665143030.$$

Thus random-function theory predicts approximately

$$0.1665143030 N \approx 10912.68$$

distinct represented states, or 16.6514% of the state space.

The committed Java and Rust experiments report median exact Hellman coverage near 16.8%. The purpose of the larger sweep is to determine whether this agreement persists across state dimensions and chain shapes, not merely at one configuration.

For contrast, Eq. (3) gives

$$1 - e^{-1} \approx 63.2121\%$$

because $mt = N$. The difference between 63.2% and 16.65% is therefore not itself a novel empirical anomaly; it is what established fixed-map Hellman coverage theory already leads us to expect.

4.4 The matrix-stopping parameter

Hellman’s standard parameterization is often discussed under the *matrix-stopping rule*

$$mt^2 \approx N, \tag{7}$$

rather than $mt \approx N$ [Hel80, MH09]. Under the exact condition $mt^2 = N$, Eq. (4) becomes

$$s_{k+1} = 1 - \exp(-1/t^2) \exp(-s_k),$$

and

$$\text{ECR}(t) = t s_t.$$

Our initial configuration instead satisfies

$$mt = N, \quad mt^2 = 64N.$$

It is therefore a useful high-overlap finite-state stress point, but it should not be described as a standard matrix-stopping configuration. The experiment matrix includes both regimes:

1. configurations near $mt^2 = N$, to connect directly with classical parameterization;
2. equal- mt chain-shape sweeps, to isolate the effect of changing the number and length of rows at fixed nominal transition budget;
3. the historical/repository stress configuration $m = 1024, t = 64$ at $N = 2^{16}$.

4.5 Finite- N and model limitations

Equation (4) is an approximation derived under a random-function model. It is not an exact identity for an arbitrary fixed mapping. Two sources of deviation are therefore relevant to the present study.

First, the reduced state spaces are finite enough that the entire functional graph can be enumerated. Small- N corrections, self-intersections, and specific component structure need not equal their asymptotic expectations. Ma and Hong themselves discuss a special discrepancy for the $m = 1$ case and give a separate exact expression [MH09].

Second, F_b is not sampled uniformly from the set of all functions $X_b \rightarrow X_b$: it is built from DES encryption, a fixed plaintext, a structured key embedding, and a deterministic reduction. Agreement with Eq. (6) is therefore an empirical property to be tested, not an assumption.

These observations motivate a direct random-map control. For every state dimension and TMTO parameter set used in the final experiments, we will compare the DES-derived map with independently generated random mappings

$$F_{\text{rand}} : X_b \longrightarrow X_b$$

under the same start states and table parameters.

4.6 Distinguished-point baseline

For the predicate

$$D_d(x) = 1 \iff x \bmod 2^d = 0,$$

exactly $N/2^d$ states are distinguished. If successive states behaved as fresh uniform samples, the first distinguished hit would have geometric parameter

$$p = 2^{-d}.$$

For an unbounded chain,

$$\Pr[\tau = k] = (1 - p)^{k-1}p, \quad \mathbb{E}[\tau] = 2^d.$$

With a hard transition bound L , the fresh-sample truncation probability is

$$\Pr[\tau > L] = (1 - 2^{-d})^L. \tag{8}$$

This is again a baseline rather than an exact statement for iterations of a fixed finite mapping. Cycles that contain no distinguished state can make a row fail to terminate even when L is large, and merged trajectories create dependence among rows. The final experiments compare measured truncation and coverage with Eq. (8) and with exact functional-graph structure.

4.7 What constitutes a meaningful deviation

For each configuration we compute the theory residual

$$\Delta_{\text{cov}} = \rho_{\text{measured}} - \rho_{\text{MH}}.$$

A single nonzero residual is not evidence of cryptographic structure: finite-size variability and the approximation itself must be accounted for. The final paper therefore reports distributions over repeated random start sets and, for the random-map controls, over independently sampled mappings. Only a deviation that is stable relative to these controls will be interpreted as evidence that the DES-derived map/reduction has coverage behavior not captured by the random-function model.

5 Reference Implementations

The artifact contains two independently written implementations of the finite state model in Section 3. Their purpose is not merely to provide two programming-language front ends. Agreement between independent implementations is used as evidence that the observed table statistics follow from the specified TMTO semantics rather than accidental behavior of one code base.

5.1 Java 17 reference

The Java implementation uses the standard Java cryptography interface for `DES/ECB/PKCS5Padding`. The reduced-state layer constructs the eight-byte DES key explicitly, inserts odd parity in every byte, and varies only the effective data bits defined in Section 3.

The core components are `ReducedDesKeySpace`, `DesOracle`, `LegacyReduction`, `HellmanTable`, `DistinguishedPointTable`, and `CoverageAnalyzer`. Endpoint collisions are retained, and online lookup always verifies the full target ciphertext.

5.2 Rust reference

The Rust crate implements the same semantics independently. It uses the RustCrypto `des` crate for the DES block primitive and explicitly performs PKCS#5-compatible padding before ECB-style block encryption. State encoding, parity, reduction, Hellman construction, distinguished-point construction, candidate regeneration, and exact coverage are separately implemented in Rust.

The independence matters because equality requires agreement on byte order, padding, parity, reduction bit positions, coverage conventions, truncated DP rows, endpoint collisions, and candidate verification.

5.3 Literal shared experiment plan

Small committed vectors already compare state encoding, key bytes, ciphertexts, and reductions. For table-level equivalence we use the stronger workload

`experiments/plans/paper-b16-shared/`.

It contains literal CSV files specifying every target, every Hellman start, every distinguished-point start, and all common parameters.

The canonical workload uses

$$b = 16, \quad N = 65536, \quad m_H = 1024, \quad t_H = 64,$$

and

$$m_D = 1024, \quad d = 6, \quad L = 256$$

over 30 trials.

The literal plan removes a confounder present in the first benchmark runs. The Java harness generated trials with `SplittableRandom`, while the first Rust harness used `SplitMix64`. Both runs were deterministic, but they were not identical workloads. The committed CSV files now define the workload; neither language’s PRNG participates in the equivalence experiment.

5.4 Algorithmic equivalence output

For every trial both runners emit deterministic fields: target state and ciphertext, exact Hellman coverage, distinct endpoints, recovery state, endpoint matches and regenerated candidates, as well as exact DP coverage, stored/truncated rows, endpoint counts, recovery state, endpoint candidates, candidate rows, and projection steps.

No timing field is included. The Java and Rust semantic CSV files are required to be byte-for-byte identical before a cross-language timing experiment is admitted into the paper.

5.5 Separation of correctness and timing

The experimental order is

semantic equivalence $\longrightarrow$ identical workload $\longrightarrow$ performance comparison.

Exact coverage is also computed outside timed preprocessing and lookup paths. The current section treats deterministic Java/Rust equality as correctness evidence; performance is considered only after that gate.

6 Experimental Methodology

The experiments are designed to separate three questions that are easy to conflate: whether the implementations agree, whether the concrete DES-derived mapping follows random-function TMTO theory, and how wall-clock performance depends on implementation language. The first question is resolved before the second is measured, and both are resolved before the third is interpreted.

6.1 Semantic-equivalence gate

Section 5 describes the committed 30-trial literal workload consumed by Java and Rust. The resulting deterministic CSV files are required to be byte-for-byte identical. Timing fields are deliberately absent from this gate. Thus any later performance comparison starts from identical targets, starts, table parameters, ciphertexts, coverage, endpoint counts, and recovery decisions.

6.2 Exact-map experiment rather than a timing benchmark

The coverage experiment is not timed cryptographic benchmarking. For each

$$b \in \{12, 14, 16, 18\},$$

the Rust experiment harness first enumerates the entire concrete mapping

$$F_b : X_b \rightarrow X_b$$

once and stores its next-state array. Every later table traversal for that state dimension is then performed over this exact array. This has two advantages: coverage experiments do not repeatedly pay DES implementation cost, and the resulting measurements depend only on the finite functional graph and the selected starts.

The complete mapping is feasible to enumerate at these dimensions: the largest space used in this paper has only

$$2^{18} = 262144$$

vertices.

6.3 Random-mapping control

For every state dimension and trial we also generate one independent pseudo-random control mapping

$$F_{\text{rand}} : X_b \rightarrow X_b$$

by assigning every vertex an independent SplitMix64-generated output reduced modulo N . SplitMix64 is used only as a deterministic experiment generator; no cryptographic pseudorandomness claim is required.

Within a fixed trial and parameter configuration, the DES-derived mapping and the random control receive exactly the same sorted set of start states. Consequently,

$$\Delta_{\text{map}} = \rho_{\text{DES}} - \rho_{\text{rand}}$$

does not contain start-set variation for that trial.

The random mapping changes across trials. The DES-derived F_b is fixed for a given b , while its selected start set changes. This distinction is intentional: the random controls estimate the variability across random functions, whereas the DES measurements characterize repeated table constructions on one concrete cryptographic functional graph.

6.4 Hellman parameter families

We use two complementary parameter families. The first consists of matrix-stopping configurations satisfying

$$mt^2 = N.$$

The second holds the nominal represented-position budget fixed at

$$mt = N$$

for $b = 16$ while varying the chain shape. The original repository stress configuration

$$(m, t) = (1024, 64)$$

is included in this second family.

The theory-only configurations and their Ma-Hong predictions are generated from committed code before measurements are run. The resulting grid is:

Configuration	b	m	t	mt/N	ρ_{MH}
stop-b12-a	12	64	8	0.1250	10.444%
stop-b12-b	12	16	16	0.0625	5.303%
stop-b14-a	14	64	16	0.0625	5.303%
stop-b14-b	14	16	32	0.0312	2.671%
stop-b16-a	16	256	16	0.0625	5.303%
stop-b16-b	16	64	32	0.0312	2.671%
stop-b16-c	16	16	64	0.0156	1.341%
stop-b18-a	18	256	32	0.0312	2.671%
stop-b18-b	18	64	64	0.0156	1.341%
stop-b18-c	18	16	128	0.0078	0.671%
budget-b16-t16	16	4096	16	1.0000	31.169%
budget-b16-t32	16	2048	32	1.0000	22.951%
stress-b16-current	16	1024	64	1.0000	16.651%
budget-b16-t128	16	512	128	1.0000	11.985%

Table 1: Hellman parameter configurations. The last column is the Ma-Hong random-function prediction for the fraction of the whole state space covered by the first t columns.

6.5 Repeated trials and deterministic seeds

Each configuration is evaluated for 30 trials on the DES-derived mapping and 30 matched random-function controls. Start sets are distinct within a table and are generated deterministically from a documented base seed, state dimension, trial index, and configuration index. The seed derivation is part of the committed experiment source. Re-running the same commit therefore reconstructs the same workloads and random maps.

The shared Java/Rust equivalence plan is independent of this sweep generator. Its literal CSV files remain the canonical evidence for cross-language semantic equality.

6.6 Primary Hellman observables

For each table we record:

$$\begin{aligned} C_H &= \text{number of distinct represented states,} \\ \rho_H &= C_H/N, \\ E_H &= \text{number of distinct endpoints,} \\ D_H &= mt - C_H. \end{aligned}$$

The quantity D_H is a compact count of repeated nominal matrix positions; it does not distinguish self-intersections from merges between different rows.

For every configuration we also record the theory residual

$$\Delta_{\text{MH}} = \rho_H - \rho_{\text{MH}}(N, m, t),$$

where ρ_{MH} is computed from Eq. (6). The independent occupancy quantity

$$\rho_{\text{iid}} = 1 - e^{-mt/N}$$

is retained only as a reference column, not as the Hellman null model.

6.7 Interpretation rule

A close match between one DES-derived table and the Ma-Hong recurrence is not by itself evidence that the concrete mapping behaves randomly. The paper therefore compares distributions across start sets with distributions across independently generated random maps. We interpret a DES-vs-theory residual as structurally interesting only if it is stable relative to the random-map control distribution.

Similarly, changing t at fixed mt simultaneously changes row count, chain length, endpoint structure, and the amount of permanent coalescence. Results from that family are interpreted as chain-shape effects rather than as differences in nominal precomputation budget.

6.8 Distinguished-point experiments

The distinguished-point sweep fixes

$$b = 16, \quad N = 65536, \quad m = 1024$$

and varies

$$d \in \{4, 6, 8\}.$$

For each d , the nominal fresh-sample mean stopping time is 2^d . We use three hard bounds,

$$L \in \{2^d, 2^{d+1}, 2^{d+2}\},$$

so that $L/\mathbb{E}[\tau] \in \{1, 2, 4\}$ under the geometric baseline.

Each of the nine (d, L) configurations is evaluated for 30 trials on the fixed DES-derived mapping and on matched independently generated random mappings. Within a trial, all nine configurations and both map types use the same set of 1024 distinct starts. We record exact coverage of stored rows, stored and truncated row counts, truncation fraction, distinct distinguished endpoints, and mean transition count among stored rows.

The fresh-sample truncation reference

$$(1 - 2^{-d})^L$$

is reported beside both map types. Because fixed functional graphs can enter cycles containing no distinguished state, it is treated as a reference rather than as an exact finite-map prediction.

7 Results

7.1 Cross-language semantic equivalence

Java and Rust consumed the same literal 30-trial workload containing 30 targets, 30,720 Hellman starts, and 30,720 distinguished-point starts. The complete deterministic CSV outputs were byte-for-byte identical, including target ciphertexts, exact coverage, endpoint counts, recovery decisions, and lookup counters.

On that workload, Hellman recovered 4/30 targets and had median exact coverage 16.825%. The distinguished-point table recovered 3/30 targets and had median exact coverage 15.412%. These are correctness results; no cross-language timing claim is inferred.

7.2 Hellman coverage tracks the random-map control

Table 2 reports the $b = 16$ configurations. Across both the matrix-stopping family and the equal- mt family, the DES-derived map tracks the matched random-map control closely.

Configuration	m	t	Ma–Hong	DES median	Random median
stop-b16-a	256	16	5.303%	5.319%	5.302%
stop-b16-b	64	32	2.671%	2.685%	2.654%
stop-b16-c	16	64	1.341%	1.357%	1.350%
budget-b16-t16	4096	16	31.169%	31.818%	31.720%
budget-b16-t32	2048	32	22.951%	23.180%	23.066%
stress-b16-current	1024	64	16.651%	16.840%	16.810%
budget-b16-t128	512	128	11.985%	12.158%	11.987%

Table 2: Measured median exact Hellman coverage over 30 trials for the DES-derived mapping and matched random mappings. Ma–Hong is the random-function prediction from Eq. (6).

At the original stress point

$$N = 65536, \quad m = 1024, \quad t = 64,$$

the Ma–Hong prediction is 16.651%, while the measured medians are 16.840% for the DES-derived map and 16.810% for the random-map controls. The mean paired DES-minus-random difference is 0.087 percentage points. A deterministic nonparametric bootstrap over the 30 paired trials gives a descriptive 95% interval of $[-0.071, 0.244]$ percentage points.

Both the concrete map and random controls exhibit the large loss relative to independent occupancy, while both remain close to the established fixed-map Hellman baseline.

7.3 Finite-size residuals are shared by random controls

The Ma–Hong recurrence is asymptotic, so small residuals are expected at finite N . In the sweep, residuals of similar scale also appear in the random-map controls. The configuration with the largest absolute paired mean DES-minus-random difference is **budget-b16-t128**, at 0.218 percentage points.

Within the tested dimensions and parameters, this provides no evidence that the DES-derived mapping has coverage behavior qualitatively different from the finite random mappings.

7.4 Chain shape matters at fixed nominal budget

The equal-budget $b = 16$ family fixes $mt = N$ while varying t . Table 2 shows that coverage is not determined by mt alone. Shorter rows and more starts cover a larger fraction of the state space; longer rows undergo more permanent coalescence under the fixed map. Both DES and random-map measurements follow the trend predicted by the Ma–Hong recurrence.

7.5 Distinguished-point truncation

The distinguished-point sweep fixes $b = 16$ and $m = 1024$, varies $d \in \{4, 6, 8\}$, and uses hard bounds equal to one, two, or four times the fresh-sample mean 2^d .

d	L	Fresh trunc.	DES trunc.	Random trunc.	DES coverage
4	16	35.61%	34.33%	34.52%	5.78%
4	32	12.68%	12.40%	12.06%	9.61%
4	64	1.61%	0.93%	1.42%	11.77%
6	64	36.50%	33.98%	36.47%	9.50%
6	128	13.32%	8.35%	15.23%	14.06%
6	256	1.77%	0.29%	5.18%	15.49%
8	256	36.72%	31.64%	39.26%	11.14%
8	512	13.48%	29.69%	32.23%	11.49%
8	1024	1.82%	29.69%	32.23%	11.49%

Table 3: Distinguished-point results over 30 trials. “Fresh trunc.” is $(1 - 2^{-d})^L$; DES and random truncation columns are measured medians. Coverage is the exact fraction represented by stored DES-derived rows, with truncated rows excluded according to Section 3.

For $d = 6$ and $L = 256$, the fresh-sample truncation reference is 1.77%. The measured median truncation rates are 0.29% for the DES-derived map and 5.18% for the random-map controls, while median exact DES coverage is 15.49%.

The geometric expression is only a fresh-sample reference. Fixed finite mappings can merge trajectories or enter cycles without a distinguished vertex, so the matched random-map controls provide the more relevant finite-functional-graph comparison.

7.6 Exact explanation of the DP truncation plateau

The bounded-start experiment samples only 1024 starts per trial. Because $N = 65536$ is still enumerable, we can remove this sampling layer and compute the stopping behavior for *every* possible start state.

For a fixed map and distinguished set D , reverse reachability from D classifies each vertex according to whether its forward trajectory can ever reach a distinguished state. Since our chain convention tests $F(s)$ after the first transition, the eventual truncation floor is the fraction of starts s for which $F(s)$ lies outside the reverse-reachable set.

This computation explains the apparent long-bound plateau in Table 3. For $d = 8$, for example, the DES-derived map has an exact eventual truncation floor of 29.71%, compared with a median 33.27% over the 30 random mappings. Increasing L cannot recover starts whose trajectories enter components whose eventual cycle contains no distinguished state.

The same effect is smaller for $d = 4$ and $d = 6$: the exact DES floors are 0.03% and 0.31%, respectively. Thus the fresh-sample geometric model is a useful short-chain reference, while the whole functional graph determines the nonzero large- L limit.

d	L	DES exact trunc.	Random median	DES eventual floor
4	16	34.38%	34.86%	0.03%
4	32	12.32%	11.84%	0.03%
4	64	0.95%	1.38%	0.03%
6	64	33.64%	35.77%	0.31%
6	128	8.50%	14.15%	0.31%
6	256	0.31%	5.06%	0.31%
8	256	31.69%	38.26%	29.71%
8	512	29.71%	33.27%	29.71%
8	1024	29.71%	33.27%	29.71%

Table 4: Exact whole-state distinguished-point truncation for the DES-derived map and medians over 30 complete random mappings. The DES eventual floor is the fraction of starts that can never reach a distinguished point after the first transition, irrespective of the chain bound.

7.7 What the experiments establish

Within the reduced state spaces tested here, the evidence supports three specific claims. First, Java and Rust agree on the full deterministic semantics of the committed shared workload. Second, exact Hellman coverage of the DES-derived mappings is close to both the Ma–Hong random-function prediction and matched finite random-map controls. Third, distinguished-point truncation and coverage can be measured exactly and compared with both fresh-sample and finite-map references.

These claims are deliberately narrower than saying that DES itself is a random function. The tested object is the reduced mapping F_b induced by the chosen key embedding, fixed plaintext, DES primitive, and reduction.

8 Discussion and Limitations

8.1 What the Hellman experiment says

The Hellman sweep began from an apparent discrepancy: at the configuration $mt = N$, an independent-occupancy heuristic suggests about 63.2% coverage, while the original experiment measured only about 16.8%. Once the correct fixed-map baseline is used, the discrepancy disappears. The Ma–Hong recurrence predicts 16.651%, and both the DES-derived mapping and matched random maps are close to that value.

This is a useful negative result. It prevents an implementation artifact or an inappropriate iid approximation from being promoted into a cryptanalytic claim. Within the state dimensions tested here, we find no stable Hellman coverage anomaly that distinguishes the DES-derived mapping from the random controls.

8.2 What the distinguished-point experiment adds

Distinguished points expose a different finite-functional-graph effect. The fresh-sample model predicts an exponentially decreasing truncation probability $(1 - 2^{-d})^L$. For a fixed mapping, however, a trajectory eventually enters a cycle. If the relevant eventual cycle does not contain a distinguished state, increasing L beyond the transient and cycle lengths cannot make the chain terminate.

The exact reverse-reachability computation quantifies this phenomenon over all 2^{16} starts. In particular, the large- L truncation plateau observed for $d = 8$ is not a timing artifact or a failure

of the DP implementation: it is a property of the complete functional graph. The classical DP literature already treats bounded chains and cycle removal as practical concerns [SRQL03, AJO08]; our contribution here is the exact executable finite-state accounting for the concrete map and matched random controls.

8.3 Why reduced DES is useful here

DES is obsolete and the experiments do not attempt to recover a full DES key. The reduced model is useful precisely because exact enumeration is possible. For $b \leq 18$ we can construct the entire iteration map, measure exact Hellman coverage, and compare with complete random mappings. At $b = 16$ we can also compute exact reverse reachability for every distinguished-point predicate.

The resulting object is best viewed as a finite cryptanalytic laboratory: the primitive supplies a concrete deterministic mapping, while the reduced dimension makes functional-graph quantities observable without statistical estimation.

8.4 Limitations

Several limitations delimit the claims. First, the DES-derived mapping depends on one fixed plaintext, one key embedding, and one reduction. Different plaintexts or reductions induce different functional graphs. Second, the Hellman study stops at $b = 18$ and the exact distinguished-point graph analysis is performed at $b = 16$; no asymptotic statement about larger DES subspaces follows automatically. Third, the random controls are deterministic SplitMix64-generated mappings used for reproducibility. They are experimental stand-ins for uniformly sampled functions, not cryptographic random oracles.

Fourth, the bootstrap intervals in Section 7 are descriptive summaries of the chosen 30 paired trials. We do not present them as a formal proof of equivalence between the DES-derived map and the random-function distribution. Fifth, the Java/Rust result establishes semantic equality on a shared workload, not equality of all possible implementations or providers.

Finally, no Java-versus-Rust speedup claim appears in this version. Although both languages contain timing harnesses, the scientific contribution of this paper is exact finite-state behavior and reproducibility. A performance study requires a separately controlled benchmarking design.

8.5 Implications for TMTO experimentation

The experiments suggest three practical rules for reproducible TMTO studies: use the coverage model appropriate to iterated fixed functions rather than iid occupancy; separate exact semantic validation from performance measurement; and, for distinguished points, distinguish short-chain geometric behavior from the eventual reachability structure of the functional graph.

9 Reproducibility

The complete artifact is maintained at

<https://github.com/qoosmo/cryptanalytic-time-memory-tradeoffs>.

The paper source is under

[papers/hellman-exact-coverage/](#).

A release tag identifying the submitted ePrint snapshot is created only after the final PDF and metadata checks; that tag is the preferred long-term reproduction anchor.

9.1 Committed evidence

The repository contains:

- the Java 17 implementation and tests;
- the independent Rust implementation and tests;
- the literal Java/Rust shared experiment plan;
- byte-identical Java/Rust semantic result files;
- the 840-row Hellman DES/random-map sweep;
- the 540-row bounded distinguished-point sweep;
- the exact whole-state distinguished-point reachability analysis;
- environment metadata for the generated experiment files;
- Python scripts that transform raw CSV files into paper tables and result macros.

Generated LaTeX tables are committed, but they are not primary evidence: each is reproducible from a raw result file and analysis script.

9.2 Correctness gates

Before the paper experiments are accepted, the artifact runs the Java unit suite, Rust tests, Rust formatting, and Clippy with warnings denied. The shared-plan gate additionally requires the Java and Rust result CSV files to be byte-for-byte identical. Exact coverage diagnostics are kept outside timed paths.

9.3 Evidence pipeline

The analysis pipeline is

committed source + parameters
→ raw experiment CSV
→ analysis scripts
→ generated tables/macros
→ PDF.

No numerical value in the results section needs to be copied manually into the manuscript.

10 Conclusion

We built a reproducible finite-state laboratory for classical cryptanalytic time–memory tradeoffs and used it to separate independent occupancy, fixed-map Hellman coverage, and distinguished-point stopping on a finite functional graph.

For Hellman tables, the initially surprising 16.8% coverage at $mt = N$ is not anomalous. The established Ma–Hong random-function recurrence predicts about 16.651%, and exact measurements on the DES-derived map closely track both that prediction and matched random-map controls

across the tested grid. At fixed mt , chain shape still matters because longer iterates create more permanent coalescence.

For distinguished points, the fresh-sample geometric truncation model describes only the short-chain reference. Exact reverse reachability over all 2^{16} states shows a nonzero eventual truncation floor whenever some trajectories cannot reach the distinguished set. This directly explains the large- L plateaus observed in the bounded-chain experiments.

Finally, independent Java and Rust implementations produce byte-identical deterministic outputs on a shared 30-trial workload. The main contribution is therefore not a new generic TMTO construction, but an executable bridge between established theory, complete finite functional graphs, concrete cryptographic mappings, and independently reproducible implementations.

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